$$\text{and} \quad \lim_{x\to-\infty}\frac{2x^2-8}{x^2-16} = \lim_{x\to-\infty}\frac{2-\frac{8}{x^2}}{1-\frac{16}{x^2}} = 2$$

Therefore, $y = 2$ is a horizontal asymptote. This can also be obtained by synthetic division. ■

7.10 Sketching of Curves

When we are sketching the graph of functions either by hand or through any graphing software we cannot show the entire graph. Only a part of the graph can be sketched. Hence a crucial question is which part of the curve we need to show and how to decide that part. To decide on this we use the derivatives of functions. We enlist few guidelines for determining a good viewing rectangle for the graph of a function. They are :

(i) The domain and the range of the function.
(ii) The intercepts of the cure (if any).
(iii) Critical points of the function.
(iv) Local extrema of the function.
(v) Intervals of concavity.
(vi) Points of inflexions (if any).
(vii) Asymptotes of the curve (if exists)

Example 7.69

Sketch the curve $y = f(x) = x^2 - x - 6$.

Solution

Factorising the given function, we have
$y = f(x) = (x-3)(x+2)$.

(1) The domain of the given function $f(x)$ is the entire real line.

(2) Putting $y = 0$ we get $x = -2, 3$. Therefore the x-intercepts are $(-2, 0)$ and $(3, 0)$ putting $x = 0$ we get $y = -6$. Therefore the y-intercept is $(0, -6)$.

(3) $f'(x) = 2x - 1$ and hence the critical point of the curve occurs at $x = \frac{1}{2}$.

(4) $f''(x) = 2 > 0, \forall x$. Therefore at $x = \frac{1}{2}$ the curve has a local minimum which is $f\left(\frac{1}{2}\right) = -\frac{25}{4}$.

(5) The range of the function is $y \geq -\frac{25}{4}$

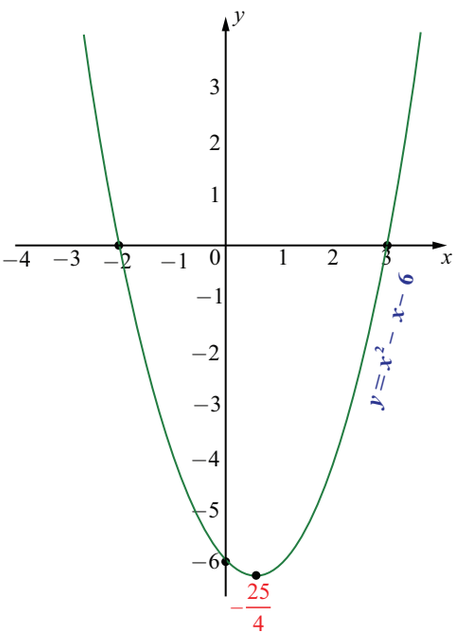


Fig.7.35

(6) Since $f''(x) = 2 > 0, \forall x$ the function is concave upward in the entire real line.

(7) Since $f(x) = 2 \neq 0, \forall x$ the curve has no points of inflection.

(8) The curve has no asymptotes.

The rough sketch of the curve is shown on the right side. ■

Chapter 7 Differential Calculus Original-new.indd 50 26-02-2020 17:27:33